

A PROOF OF TWO ENDPOINT DOUBLING BREAKS IN THE DIVISOR LISTS OF ODD COMPOSITE INTEGERS

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ABSTRACT. For a positive integer m with positive divisors $1 = d_1 < \dots < d_t = m$, let $A(m)$ be one plus the number of adjacent pairs for which the right endpoint is odd and at least twice the left endpoint. We prove that $A(m) \geq 3$ for every odd composite integer m . Consequently, the same lower bound holds for every Carmichael number. The proof is elementary: the first and last divisor adjacencies already supply two distinct qualifying pairs. The result does not classify the remaining divisor gaps or determine the sharp value of $A(m)$ for any narrower family.

1. INTRODUCTION

The ordered-divisor statistic considered here is given by a Python program in OEIS entry A237271. The same entry records the finite-range observation $a(\text{A002997}(n)) \geq 3$ for $1 \leq n \leq 10000$, where A002997 is the OEIS sequence of Carmichael numbers [1]. We give a direct proof of the corresponding universal inequality that requires neither a finite search, Korselt's criterion, nor squarefreeness. In fact, the Carmichael hypothesis can be weakened to odd compositeness.

The only Carmichael-specific fact needed is oddness, which we derive directly from the defining universal congruence. The rest uses only the least prime divisor and the correspondence $d \mapsto m/d$ on divisors.

2. DEFINITIONS AND EXACT STATEMENT

Definition 2.1. Let $m > 1$, and write the complete increasing list of its positive divisors, each occurring once, as $1 = d_1 < d_2 < \dots < d_t = m$. Define

$$A(m) = 1 + \#\{i \in \{1, \dots, t-1\} : d_{i+1} \equiv 1 \pmod{2} \text{ and } d_{i+1} \geq 2d_i\}.$$

A Carmichael number is a positive composite integer m such that $a^{m-1} \equiv 1 \pmod{m}$ for every integer a coprime to m .

The exact accepted endpoint is the following.

Theorem 2.2. *For every Carmichael number m , $A(m) \geq 3$; indeed, every odd composite integer m satisfies $A(m) \geq 3$.*

3. PROOF

Lemma 3.1. *Every Carmichael number is odd.*

Proof. Suppose that a Carmichael number m were even. Since m is composite, $m > 2$, and $\gcd(m-1, m) = 1$. Applying the defining congruence with $a = m-1$ gives $(m-1)^{m-1} \equiv 1 \pmod{m}$. But $m-1$ is odd and $m-1 \equiv -1 \pmod{m}$, so the left side is congruent to -1 . Hence m divides 2, contrary to $m > 2$. $\square$

Lemma 3.2. *Let $m > 1$ be composite, and let p be its least prime divisor. Then $d_2 = p$ and $d_{t-1} = m/p$.*

Proof. The integer p divides m . Every divisor of m greater than 1 has a prime divisor, which is at least p ; hence $d_2 = p$. If $d < m$ divides m , then m/d is an integer greater than 1 and therefore is at least p . Thus $d \leq m/p$. Since m/p is itself a proper divisor, it is the largest proper divisor and equals d_{t-1} . $\square$

Proof of Theorem 2.2. Let m be odd and composite, and let p be its least prime divisor. Then $p \geq 3$, and every divisor of m is odd. Lemma 3.2 gives the endpoint adjacencies

$$(d_1, d_2) = (1, p), \quad (d_{t-1}, d_t) = (m/p, m).$$

Both right endpoints are odd, and both ratios are $p \geq 3$, so both pairs qualify. They are distinct: if their adjacency indices coincided, then $1 = t-1$ and $t = 2$, whereas an integer with exactly two positive divisors is prime. The counted set has at least two members, and its leading 1 yields $A(m) \geq 3$. Every Carmichael number is an odd composite integer by its definition and Lemma 3.1, proving the stated consequence. $\square$

4. EXACT CERTIFICATION AND REPRODUCIBILITY

The theorem is certified by the symbolic argument above. All quantities are integers, the only quotient is m/p with $p \mid m$, and no enumerated sample is used to establish the universal claim.

During discovery, an exact-integer Python program enumerated divisors and checked examples as regression evidence. Separately, an independently written Ruby implementation used factor-pair enumeration, integer comparisons, gcd, and modular exponentiation to test all odd composite integers through 5001 and sample Carmichael numbers. It also returned $A(561) = 5$. These finite checks corroborate the endpoint lemma but are not premises of the proof; the two lemmas above are the complete exact certificate.

Related exact work on different divisor-based questions includes a certified search extension for a constraint involving the divisor-counting function [3] and an exact counterexample for a recurrence defined by the largest odd divisor [2]. These references provide methodological context only; neither studies the ordered-divisor statistic $A(m)$.

5. BOUNDARY CASES AND LIMITATIONS

Positive divisors occur once, the leading 1 is part of $A(m)$, and the comparison is non-strict. The exhibited ratios are actually $p \geq 3$, so the proof also works with $d_{i+1} > 2d_i$. Compositeness is essential to the distinctness step: for a prime the endpoint adjacencies coincide. Repeated prime factors cause no difficulty, and odd prime powers are included.

The theorem is only a lower bound. It does not enumerate all qualifying adjacencies, classify equality, determine sharp values in narrower families, or assert anything about general even composite integers. It also makes no claim about the status or novelty of other statements associated with OEIS A237271.

COMPUTATIONAL AND LANGUAGE-MODEL ASSISTANCE DISCLOSURE

Language models assisted with proof discovery, adversarial checking, manuscript drafting, and editorial revision. Exact programs were used for exploratory and regression checks, and an independently written program was used during audit. The author reviewed the definitions, scope, proof, and disclosures. The claim rests on the complete elementary proof presented here, not on model output, numerical evidence, or an unverified calculation.

REFERENCES

1. Omar E. Pol, *Entry A237271: Number of parts in the symmetric representation of $\sigma(n)$* , The On-Line Encyclopedia of Integer Sequences, 2014, Comments and programs; accessed September 4, 2026.
2. Zijian Zeng, *A counterexample to the proposed last occurrence in the largest-odd-divisor recurrence*, SSRN preprint, 2026.
3. ———, *A strict certified search extension for Erdős problem 647*, SSRN preprint, 2026.

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